

Digital Design and Computer Organization Laboratory
UE23CS251A

3rd Semester, Academic Year 2024-25

Date:22/08/2024

Name: DIPAK CHAUDHARI	PRN: PES1202402989	Section: K
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Week:___1_____

Program Number: ___1___

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT AND GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT AND WAVEFORM WITH THE AND GATE TRUTH TABLE

I. Verilog Code Screenshot

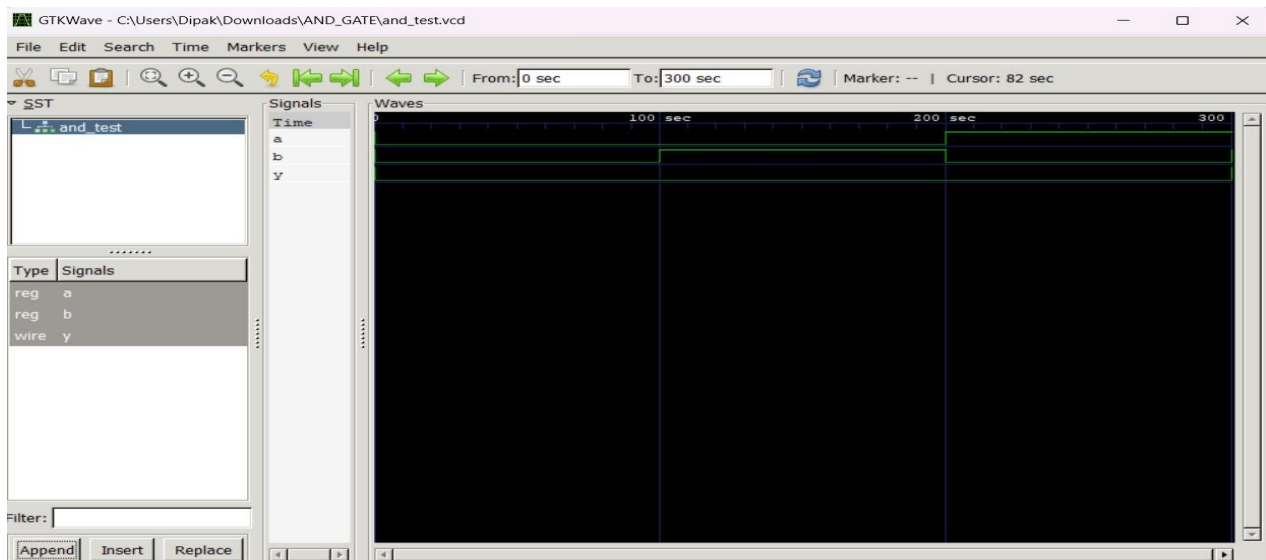
```
and_gate.v X test_bench.v and_test.vcd
and_gate.v
1 module and_gate(a,b,y);
2     input a,b;
3     output y;
4     assign y = a & b;
5 endmodule;
```

```
and_gate.v test_bench.v X and_test.vcd
test_bench.v
1 module and_test;
2
3     reg a,b;
4     wire y;
5
6     and and_gate(y,a,b);
7
8     initial begin
9         #000 a = 0; b = 0;
10        #100 a = 0; b = 1;
11        #100 a = 1; b = 0;
12        #100 a = 1; b = 1;
13    end
14
15    initial begin
16        $monitor($time," a=%b, b=%b, y=%b",a,b,y);
17    end
18
19    initial begin
20        $dumpfile("and_test.vcd");
21        $dumpvars(0,and_test);
22    end
23
24 endmodule;
```

II. Verilog VVP Output Screen Shot

```
0 a=0, b=0, y=0
100 a=0, b=1, y=0
200 a=1, b=0, y=0
300 a=1, b=1, y=1
```

III. GTKWAVE Screenshot



A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

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Week:___1_____

Program Number: ___2___

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT OR GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT OR WAVEFORM WITH THE OR GATE TRUTH TABLE

I. Verilog Code Screenshot

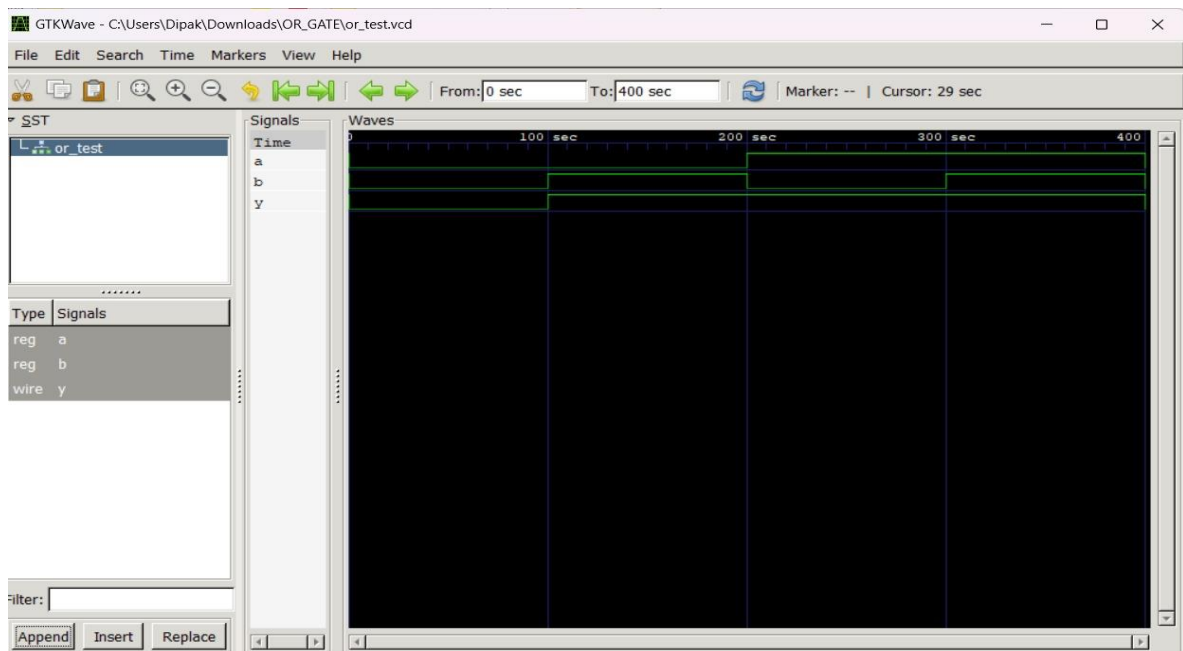
```
or_gate.v  test_bench.v
or_gate.v
1 module or_gate (a,b,y);
2     input a,b;
3     output y;
4
5     assign y = a | b;
6 endmodule
```

```
or_gate.v  test_bench.v
test_bench.v
1 module or_test;
2
3     reg a, b;
4     wire y;
5
6     // Instantiate the OR gate
7     or or_gate(y, a, b);
8
9     initial begin
10        #100 a = 0; b = 1;
11        #100 a = 1; b = 0;
12        #100 a = 1; b = 1;
13        #100 a = 0; b = 0;
14    end
15
16    initial begin
17        $monitor($time, "a=%b, b=%b, y=%b", a, b, y);
18    end
19
20    initial begin
21        // Create a VCD file for waveform simulation
22        $dumpfile("or_test.vcd");
23        $dumpvars(0, or_test);
24    end
25
26 endmodule
```

II. Verilog VVP Output Screen Shot

```
0a=0, b=0, y=0
100a=0, b=1, y=1
200a=1, b=0, y=1
300a=1, b=1, y=1
400a=0, b=0, y=0
```

III. GTKWAVE Screenshot



A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

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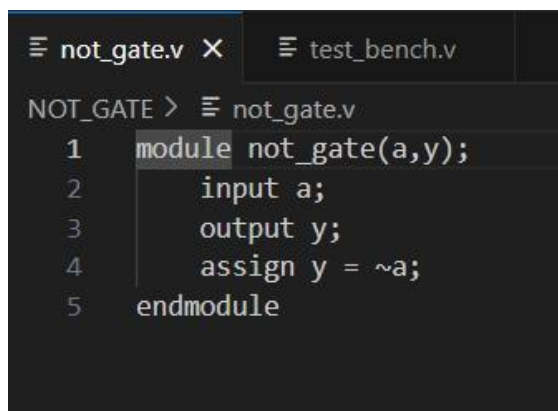
Week:___1_____

Program Number: ___3___

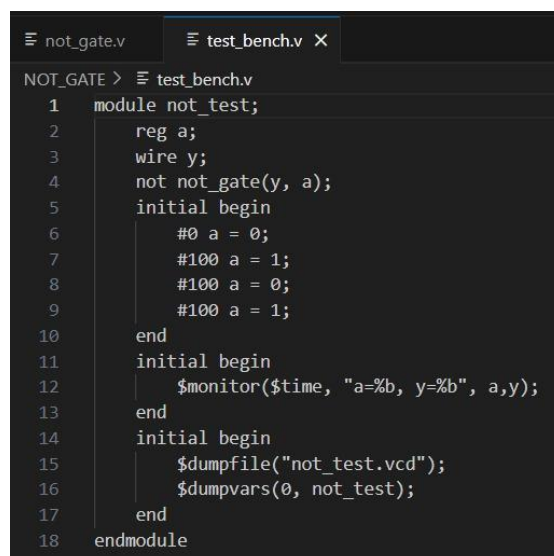
TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT NOT GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT NOT WAVEFORM WITH THE NOT GATE TRUTH TABLE

I. Verilog Code Screenshot



```
not_gate.v X test_bench.v
NOT_GATE > not_gate.v
1 module not_gate(a,y);
2     input a;
3     output y;
4     assign y = ~a;
5 endmodule
```

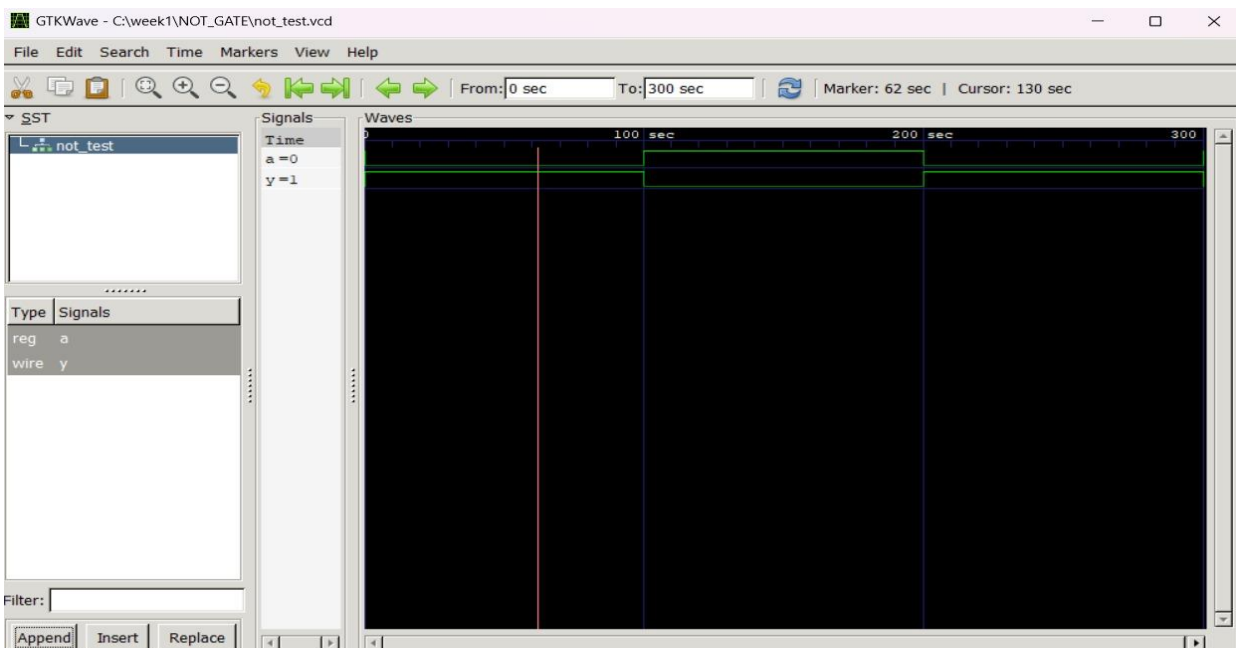


```
not_gate.v test_bench.v X
NOT_GATE > test_bench.v
1 module not_test;
2     reg a;
3     wire y;
4     not not_gate(y, a);
5     initial begin
6         #0 a = 0;
7         #100 a = 1;
8         #100 a = 0;
9         #100 a = 1;
10    end
11    initial begin
12        $monitor($time, "a=%b, y=%b", a,y);
13    end
14    initial begin
15        $dumpfile("not_test.vcd");
16        $dumpvars(0, not_test);
17    end
18 endmodule
```

II. Verilog VVP Output Screen Shot

```
0a = 0, y = 1
5a = 1, y = 0
```

III. GTKWAVE Screenshot



A	B
0	1
1	0

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Date:

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Week: 1

Program Number: 4

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT XNOR GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT NAND WAVEFORM WITH THE XNOR GATE TRUTH TABLE

I. Verilog Code Screenshot

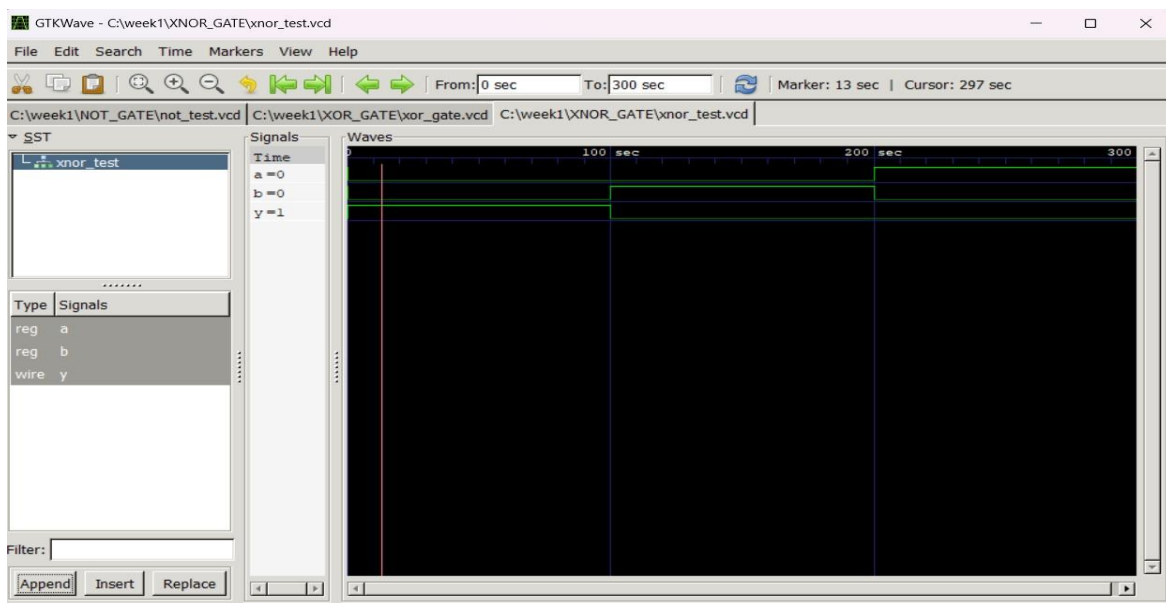
```
test_bench.v xnor_gate.v X
XNOR_GATE > xnor_gate.v
1 module xnor_gate (a, b, y);
2     input a, b;
3     output y;
4
5     assign y = ~(a ^ b);
6 endmodule
```

```
test_bench.v xnor_gate.v
XNOR_GATE > test_bench.v
1 module xnor_test;
2
3     reg a, b;
4     wire y;
5
6     // Instantiate the OR gate
7     xnor xnor_gate(y, a, b);
8
9     initial begin
10         #000 a = 0; b = 0;
11         #100 a = 0; b = 1;
12         #100 a = 1; b = 0;
13         #100 a = 1; b = 1;
14     end
15
16     initial begin
17         $monitor($time, "a=%b, b=%b, y=%b", a, b, y);
18     end
19
20     initial begin
21         // Create a VCD file for waveform simulation
22         $dumpfile("xnor_test.vcd");
23         $dumpvars(0, xnor_test);
24     end
25
26 endmodule
```

II. Verilog VVP Output Screen Shot

```
0a = 0, b = 0, y = 1
100a = 0, b = 1, y = 1
200a = 1, b = 0, y = 1
300a = 1, b = 1, y = 0
```

III. GTKWAVE Screenshot



A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

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Date:

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Week: 1

Program Number: 5

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT NOR GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT NOR WAVEFORM WITH THE NOR GATE TRUTH TABLE

I. Verilog Code Screenshot

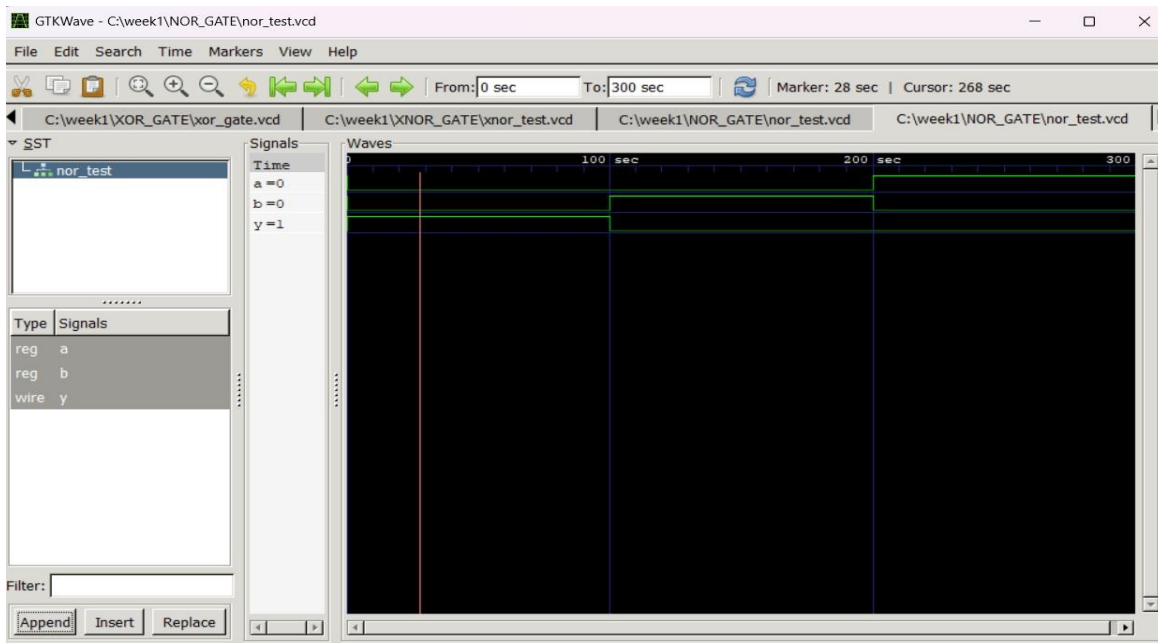
```
nor_gate.v X test_bench.v
NOR_GATE > nor_gate.v
1 module nor_gate (a,b,y);
2     input a,b;
3     output y;
4
5     assign y = ~(a | b);
6 endmodule
```

```
nor_gate.v test_bench.v X
NOR_GATE > test_bench.v
1 module nor_test;
2
3     reg a, b;
4     wire y;
5
6     // Instantiate the OR gate
7     nor nor_gate(y, a, b);
8
9     initial begin
10         #000 a = 0; b = 0;
11         #100 a = 0; b = 1;
12         #100 a = 1; b = 0;
13         #100 a = 1; b = 1;
14     end
15
16     initial begin
17         $monitor($time, "a=%b,b=%b,y=%b", a,b,y);
18     end
19
20     initial begin
21         $dumpfile("nor_test.vcd");
22         $dumpvars(0, nor_test);
23     end
24
25 endmodule
```

III. Verilog VVP Output Screen Shot

```
0a=0,b=0,y=1
100a=0,b=1,y=0
200a=1,b=0,y=0
300a=1,b=1,y=0
```

III. GTKWAVE Screenshot



A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

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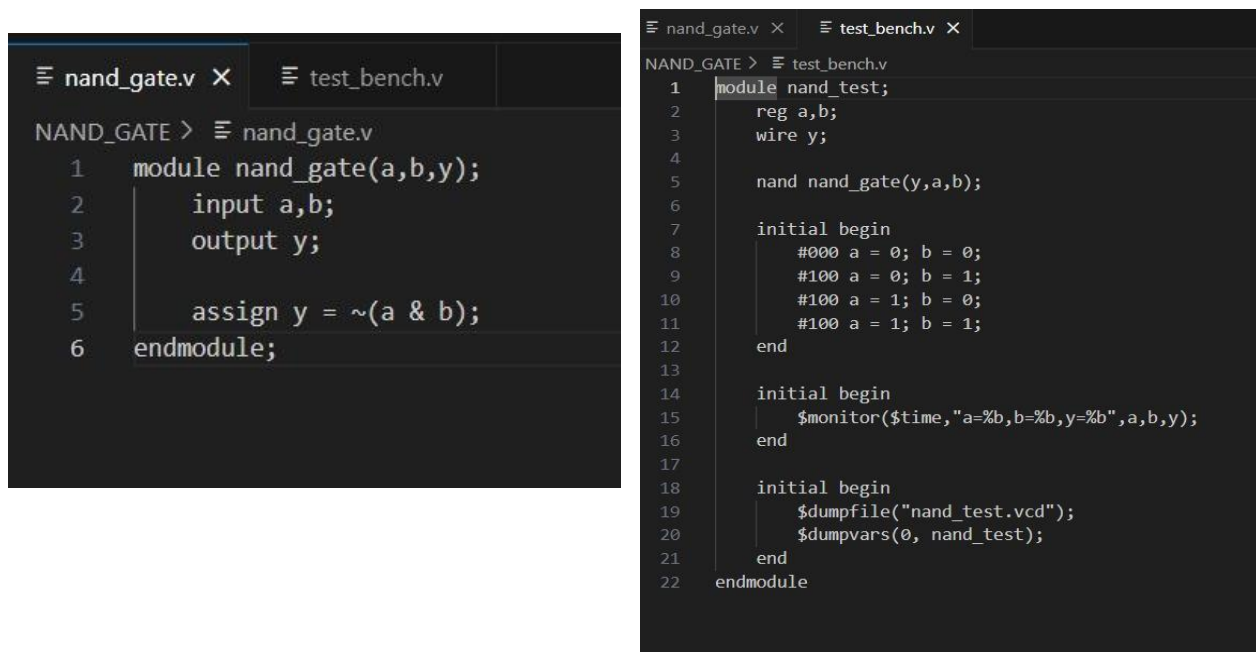
Week: 1

Program Number: 6

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT NAND GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT XOR WAVEFORM WITH THE NAND GATE TRUTH TABLE

I. Verilog Code Screenshot



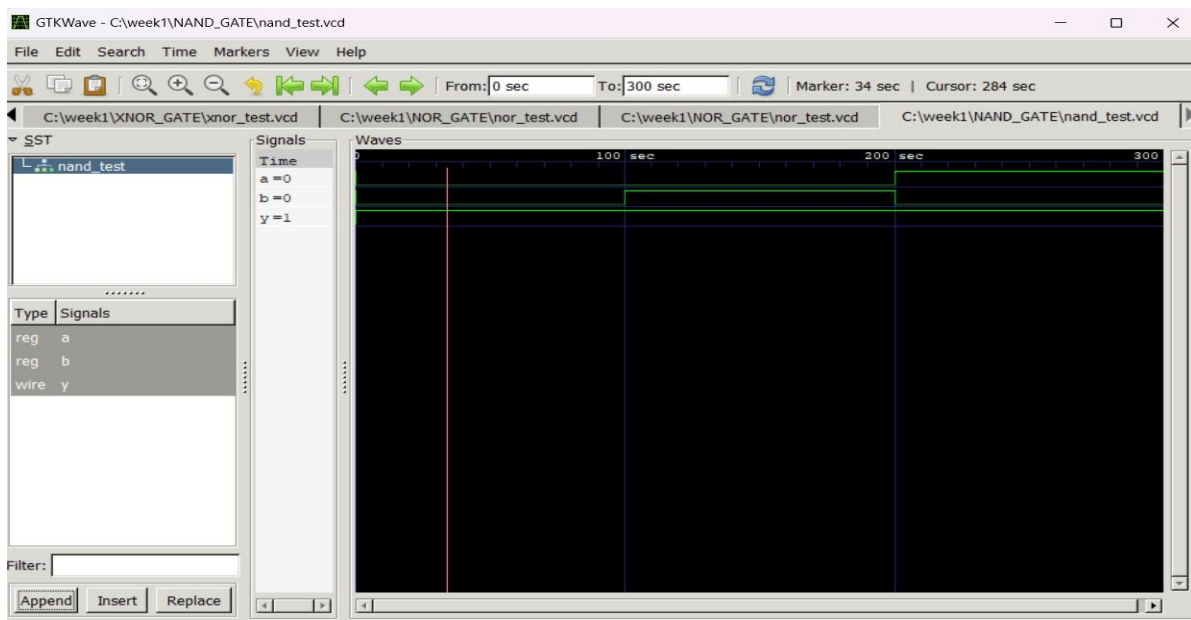
```
≡ nand_gate.v X  ≡ test_bench.v X
NAND_GATE > ≡ nand_gate.v
1  module nand_gate(a,b,y);
2      input a,b;
3      output y;
4
5      assign y = ~(a & b);
6  endmodule;

≡ nand_gate.v X  ≡ test_bench.v X
NAND_GATE > ≡ test_bench.v
1  module nand_test;
2      reg a,b;
3      wire y;
4
5      nand nand_gate(y,a,b);
6
7      initial begin
8          #000 a = 0; b = 0;
9          #100 a = 0; b = 1;
10         #100 a = 1; b = 0;
11         #100 a = 1; b = 1;
12     end
13
14     initial begin
15         $monitor($time,"a=%b,b=%b,y=%b",a,b,y);
16     end
17
18     initial begin
19         $dumpfile("nand_test.vcd");
20         $dumpvars(0, nand_test);
21     end
22 endmodule
```

II. Verilog VVP Output Screen Shot

```
0a = 0, b = 0, y = 0
5a = 0, b = 1, y = 1
15a = 1, b = 0, y = 1
30a = 1, b = 1, y = 0
```

III. GTKWAVE Screenshot



A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

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3rd Semester, Academic Year 2024-25

Date:

Name: DIPAK CHAUDHARI	PRN: PES1202402989	Section: K
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Week# 1

Program Number: 7

TITLE:

WRITE A VERILOG PROGRAM TO MODEL A TWO INPUT XOR GATE. GENERATE THE VVP OUTPUT AND SIMULATION WAVEFORM USING GTKWAVE. VERIFY THE OUTPUT XNOR WAVEFORM WITH THE XOR GATE TRUTH TABLE

I. Verilog Code Screenshot

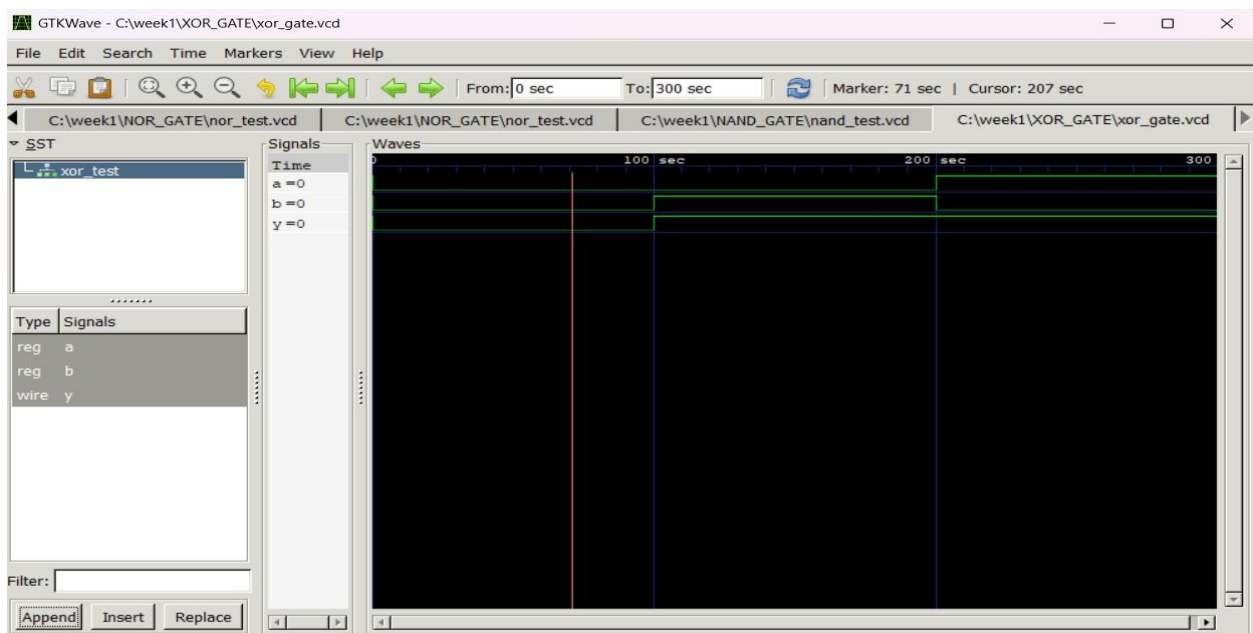
```
test_bench.v  xor_gate.v X
XOR_GATE > xor_gate.v
1  module xor_gate(a,b,y);
2      input a,b;
3      output y;
4
5      assign y = a ^ b;
6  endmodule
```

```
test_bench.v X  xor_gate.v
XOR_GATE > test_bench.v
1  module xor_test;
2      reg a, b;
3      wire y;
4
5      xor xor_gate(y, a, b);
6
7      initial begin
8          #000 a = 0; b = 0;
9          #100 a = 0; b = 1;
10         #100 a = 1; b = 0;
11         #100 a = 1; b = 1;
12     end
13
14     initial begin
15         $monitor($time, " a=%b, b=%b, y=%b", a, b, y);
16     end
17     initial begin
18         $dumpfile("xor_gate.vcd");
19         $dumpvars(0, xor_test);
20     end
21 endmodule
```

II. Verilog VVP Output Screen Shot

```
0a=0, b=0, y=1
100a=0, b=1, y=0
200a=1, b=0, y=0
300a=1, b=1, y=1
```

III. GTKWAVE Screenshot



A	B	C
0	0	0
0	1	1
1	0	1
1	1	0